

Interference Analysis for Coexistence of UAVs and Civil Aircrafts Based on Automatic Dependent Surveillance-Broadcast

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Abstract—Due to the advantages of high mobility and easy deployment, unmanned aerial vehicles (UAVs) are widely applied in both military and civilian fields. In order to strengthen the flight surveillance of UAVs and guarantee the airspace safety, UAVs can be equipped with the automatic dependent surveillance-broadcast (ADS-B) system, which periodically sends flight information to other aircrafts and ground stations (GSs). However, due to the limited resource of channel capacity, UAVs equipped with ADS-B results in the interference between UAVs and civil aircrafts (CAs), which further impacts the accuracy of received information at GSs. In detail, the channel capacity is mainly affected by the density of aircrafts and the transmitting power of ADS-B. Hence, based on the three-dimensional poisson point process, this work leverages the stochastic geometry theory to build a model of the coexistence of UAVs and CAs and analyze the interference performance of ADS-B monitoring system. From simulation results, we reveal the effects of transmitting power, density, threshold and pathloss on the performance of the ADS-B monitoring system. Besides, we provide the suggested transmitting power and density for the safe coexistence of UAVs and CAs.

Index Terms—UAV, ADS-B, civil aviation, interference analysis, poisson point process, stochastic geometry.

I. INTRODUCTION

As low-altitude aerial technology advances, unmanned aerial vehicles (UAVs) play increasingly significant roles across various domains such as collaborative reconnaissance, precision agriculture, disaster rescue, and environmental monitoring [1]. Besides, the growing multitude applications necessitate abundant UAVs, raising concerns about the

flight safety. Furthermore, the absence of on-board pilots poses potential risks to the safe operation of civil aircrafts (CAs) [2]. Hence, enhancing the airspace management for UAVs and guaranteeing the flight safety emerge as crucial imperatives [3].

In order to obtain exact aerial location information, UAVs can be equipped with automatic dependent surveillance-broadcast (ADS-B) systems [4]. Working at 1090 MHz, ADS-B is beneficial to both the flight safety and air traffic management [5]. In detail, an aircraft equipped with ADS-B can automatically broadcast its flight information to nearby aircrafts and ground stations (GSs) [6]. However, due to the limited channel capacity, if multiple UAVs utilize the same channel, the interference between UAVs and CAs cannot be neglected. Consequently, the accuracy of received information at GSs is deteriorated, which further impairs the performance of the monitoring system. In particular, the density of UAVs and the transmitting power of ADS-B are key factors on the performance of the monitoring system. As a result, this work aims to analyze the interference for the coexistence of UAVs and CAs based on ADS-B with respect to the transmitting power and the density of UAVs.

There exist a couple of related works about UAVs equipped with ADS-B. For instance, [7] points out that the performance of ADS-B system is affected by the density of UAVs, ADS-B transmitting power and the number of GSs. [8] demonstrates that the performance of the ADS-B system is affected by channel characteristics and minimum updating interval required by aircrafts. Besides, there are a few works utilizing stochastic geometry (SG) and poisson point process (PPP) to analyze UAVs networks, for example, [9] states that UAV wireless networks have natural spatial random characteristics and the channel has fading and shadowing characteristics. Therefore, SG can be utilized to analyze the performance of UAV wireless networks. [10] develops a tractable framework for signal-to-interference-plus-noise ratio (SINR) analysis in downlink heterogeneous cellular networks with flexible cell association policies. [11] compares the model of Rician channel, Rayleigh channel and Nakagami-m channel in wireless network, and analyzes the coverage rate of UAV assisted cellular network. The authors in [12] utilize SG to analyze the response delay and the successful transmission probability for a single link and a group of links based on the three-dimensional (3D) distribution of UAV swarms.

However, the above works equip UAVs with ADS-B to enhance the flight safety, but they seldom utilize SG to examine the system performance considering the interference based on ADS-B. Besides, to construct the model, we distinguish the interference taking account of the coexistence of UAVs and CAs, which is a challenging problem. In short, the contributions of this paper are summarized as follows.

- We propose the model of the coexistence of UAVs and CAs based on ADS-B techniques.
- Based on SG theory, we employ 3D-PPP to reveal the interference performance and deduce the analytic form of the received probability.
- Extensive simulations are conducted to verify the effects of transmitting power, density, threshold and pathloss on the performance of ADS-B monitoring system.

II. SYSTEM MODEL

A. Network Model

Fig. 1 depicts the system model of the coexistence of UAVs and CAs. The fixed-wing UAVs, rotary-wing UAVs and CAs are randomly

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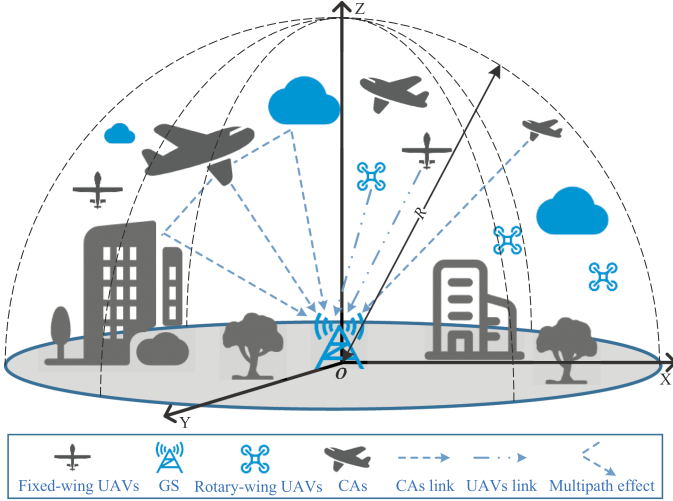


Fig. 1. System model of the coexistence of UAVs and CAs.

distributed in space $\mathbf{V}$. All aircrafts transmit flight information to the GS via ADS-B. The fixed-wing UAVs and the rotary-wing UAVs follow a 3D-PPP [13] with density λ_1 in the finite space $\mathbf{V}$, and the number of UAVs is $N_U = \lambda_1 \mathbf{V}$. The CAs follow a 3D-PPP with density λ_2 in the finite space $\mathbf{V}$, and the number of CAs is $N_C = \lambda_2 \mathbf{V}$. Denote the UAV set as $\mathcal{U} = \{U_1, \dots, U_i, \dots, U_{N_U}\}$, and the CA set as $\mathcal{C} = \{C_1, \dots, C_j, \dots, C_{N_C}\}$. It is assumed that there is only one GS in space $\mathbf{V}$, and all ADS-B packets from UAVs and CAs are received by the GS. The GS is located in the center of the ground with the coordinate of $\mathbf{O}(0, 0, 0)$. In detail, the coordinate of the i -th UAV in set $\mathcal{U}$ is $(x_{U_i}, y_{U_i}, z_{U_i})$ and the coordinate of the j -th CA in set $\mathcal{C}$ is $(x_{C_j}, y_{C_j}, z_{C_j})$. The X-axis coordinates for all aircrafts range within $[-L_x, L_x]$, the Y-axis coordinates range within $[-L_y, L_y]$, and the Z-axis coordinates are $[0, L_z]$. The euclidean distance between UAV U_i and the GS is $d_{U_i} = \sqrt{x_{U_i}^2 + y_{U_i}^2 + z_{U_i}^2}$, and the euclidean distance between CA C_j and the GS is $d_{C_j} = \sqrt{x_{C_j}^2 + y_{C_j}^2 + z_{C_j}^2}$. The transmitting power of ADS-B from UAV U_i and CA C_j are set as P_U and P_C , respectively.

B. Channel Model

G_{U_i} and G_{C_i} respectively represent the transmitter gain of UAVs and CAs. G_r denotes the receiver gain at the GS. Therefore, the total air-ground (AG) channel gain at the GS from UAVs is $G_U = G_{U_i} G_r$, and the AG channel gain between CAs and the GS is $G_C = G_{C_i} G_r$. The pathloss from the GS to UAVs and CAs are respectively proportional to $d_{U_i}^{-\alpha}$ and $d_{C_j}^{-\alpha}$, where d_{U_i} and d_{C_j} represent the distance between the aircrafts and GS. α indicates the pathloss index. h_{U_i} represents the gain of small scale fading channel between UAV U_i and the GS. h_{C_j} represents the gain of small scale fading channel between CA C_j and the GS. h_{U_i} and h_{C_j} are two random variables following an exponential distribution with mean value of 1. Gaussian white noise N is added to the model, i.e., $N = n_0 \times B$, where n_0 is noise power density and B is the system bandwidth. We leverage γ to represent SINR. Then, the γ_U^m of the desired signal sent by the m -th UAV U_m in set $\mathcal{U}$ is

$$\gamma_U^m = \frac{P_U G_U h_{U_m} d_{U_m}^{-\alpha}}{N + P_U I_{\mathcal{U} \setminus \{U_m\}} + P_C I_C}, \quad (1)$$

in which

$$I_{\mathcal{U} \setminus \{U_m\}} = \sum_{U_i \in \mathcal{U} \setminus \{U_m\}} G_U h_{U_i} d_{U_i}^{-\alpha}, \quad (2)$$

and

$$I_C = \sum_{C_j \in \mathcal{C}} G_C h_{C_j} d_{C_j}^{-\alpha}. \quad (3)$$

III. PERFORMANCE ANALYSIS

It is supposed that all the aircrafts send the flight information via ADS-B to the GS within space $\mathbf{V}$. In particular, UAVs follow the nearest neighbor association strategy [14], i.e., no other GSs outside space $\mathbf{V}$ has less distance to the target UAV

$$P\{d_{U_i} > R\} = \exp(-\lambda_1 \mathbf{V}) = \exp\left(-\frac{4}{3}\pi\lambda_1 d_{U_i}^3\right), \quad (4)$$

where $d_{U_i} \geq 0$, and R is the radius of the 3D euclidean space $\mathbb{R}^3$. Therefore, the cumulative distribution function (CDF) of the distance d_{U_i} from UAV U_i to the GS is

$$F_U(d_{U_i}) = P\{d_{U_i} \leq R\} = 1 - \exp\left(-\frac{4}{3}\pi\lambda_1 d_{U_i}^3\right), \quad (5)$$

and the probability density function (PDF) of d_{U_i} is

$$f_U(d_{U_i}) = \frac{dF_U(d_{U_i})}{d(d_{U_i})} = 4\pi\lambda_1 d_{U_i}^2 \exp\left(-\frac{4}{3}\pi\lambda_1 d_{U_i}^3\right). \quad (6)$$

The successful received probability P_{suc} at the GS is introduced to measure the transmission quality. If the distance between the UAV and GS is d , and γ is greater than the received threshold θ , the successful received probability of the GS is denoted as

$$P_{suc} = \mathbb{E}[P(\gamma \geq \theta|d)]. \quad (7)$$

Since γ is also a function of d , the P_{suc} of the m -th UAV in set $\mathcal{U}$ is further expressed as

$$\begin{aligned} P_{suc} &= \int_0^\infty P(\gamma_U^i \geq \theta|d_{U_m}) f_U(d_{U_m}) d(d_{U_m}) \\ &= \int_0^\infty 4\pi\lambda_1 d_{U_m}^2 P(\gamma_U^i \geq \theta|d_{U_m}) \\ &\quad \exp\left(-\frac{4}{3}\pi\lambda_1 d_{U_m}^3\right) d(d_{U_m}). \end{aligned} \quad (8)$$

It is assumed that the average gain of small scale fading channel is a random variable following the Gamma distribution with mean value of 1 [13], which is depicted as

$$f(h) = \frac{\beta^\beta}{\Gamma(\beta)} h^{\beta-1} e^{-\beta h}. \quad (9)$$

When β equals 1, the channel is considered as Rayleigh fading. h follows an exponential distribution with mean value of 1. The PDF of h is $f(x) = e^{-x}$, i.e., $h_{U_i} \sim \exp(1)$ and $h_{C_j} \sim \exp(1)$. Hence, $P(\gamma_U^i \geq \theta|d_{U_m})$ in (8) is further represented as

$$\begin{aligned} &P(\gamma_U^i \geq \theta|d_{U_m}) \\ &= P\left(h_{U_m} \geq \frac{\theta d_{U_m}^\alpha (N + P_U I_{\mathcal{U} \setminus \{U_m\}} + P_C I_C)}{P_U G_U}\right) \\ &= \exp\left(-\frac{\theta d_{U_m}^\alpha (N + P_U I_{\mathcal{U} \setminus \{U_m\}} + P_C I_C)}{P_U G_U}\right) \\ &= \exp\left(-\frac{\theta d_{U_m}^\alpha N}{P_U G_U}\right) \mathbb{I}_{I_{\mathcal{U} \setminus \{U_m\}}} \left(\frac{\theta d_{U_m}^\alpha}{G_U}\right) \mathbb{I}_{I_C} \left(\frac{\theta d_{U_m}^\alpha}{G_U} \times \frac{P_C}{P_U}\right). \end{aligned} \quad (10)$$

Let $\frac{\theta d_{U_m}^\alpha}{G_U} = s_1$, and we have

$$\mathbb{L}_{I_{U \setminus \{U_m\}}} \left(\frac{\theta d_{U_m}^\alpha}{G_U} \right) = \mathbb{L}_{I_{U \setminus \{U_m\}}}(s_1) = \mathbb{E}[e^{-s_1 I_{U \setminus \{U_m\}}}], \quad (11)$$

which is the Laplace transform of $I_{U \setminus \{U_m\}}$, and is further derived as

$$\begin{aligned} \mathbb{L}_{I_{U \setminus \{U_m\}}}(s_1) &= \mathbb{E} \left[\exp \left(-s_1 \sum_{U_i \in U \setminus \{U_m\}} G_U h_{U_i} d_{U_i}^{-\alpha} \right) \right] \\ &\stackrel{(a)}{=} \mathbb{E} \left[\prod_{U_i \in U \setminus \{U_m\}} \frac{1}{1 + s_1 G_U d_{U_i}^{-\alpha}} \right] \\ &\stackrel{(b)}{=} \exp \left[-\lambda_1 \int_V \left(1 - \frac{1}{1 + s_1 G_U d_{U_i}^{-\alpha}} \right) d(d_{U_i}) \right] \\ &\stackrel{(c)}{=} \exp \left[-\lambda_1 \int_{-L_x}^{L_x} \int_{-L_y}^{L_y} \int_0^{L_z} 1 - \frac{1}{1 + s_1 G_U d_{U_i}^{-\alpha}} dx dy dz \right] \\ &\stackrel{(d)}{=} \exp(-\lambda_1 H_1), \end{aligned} \quad (12)$$

where (a) is obtained by the moment generating function, (b) follows the probability generating function (PGFL) of the PPP [15], d_{U_i} in (c) can be further expressed as $\sqrt{x_{U_i}^2 + y_{U_i}^2 + z_{U_i}^2}$, and H_1 in (d) represents the triple integral in (c). Moreover, let $\frac{\theta d_{U_m}^\alpha}{G_U} \times \frac{P_C}{P_U} = s_2$, and we have

$$\mathbb{L}_{I_C} \left(\frac{\theta d_{U_m}^\alpha}{G_U} \times \frac{P_C}{P_U} \right) = \mathbb{L}_{I_C}(s_2) = \mathbb{E}[e^{-s_2 I_C}], \quad (13)$$

following the Laplace transformation of I_C . Therefore, (13) is further simplified as

$$\begin{aligned} \mathbb{L}_{I_C}(s_2) &= \mathbb{E} \left[\exp \left(-s_2 \sum_{C_j \in C} G_C h_{C_j} d_{C_j}^{-\alpha} \right) \right] \\ &= \mathbb{E} \left[\prod_{C_j \in C} \frac{1}{1 + s_2 G_C d_{C_j}^{-\alpha}} \right] \\ &\stackrel{(e)}{=} \exp(-\lambda_2 H_2), \end{aligned} \quad (14)$$

where H_2 in (e) symbolizes the triple integral. By substituting (10), (11), (12), (13) and (14) into (8), P_{suc} is calculated as

$$\begin{aligned} P_{suc} &= \int_0^\infty 4\pi\lambda_1 d_{U_m}^2 \exp \left(\frac{-\theta d_{U_m}^\alpha N}{P_U G_U} - \lambda_1 H_1 \right. \\ &\quad \left. - \lambda_2 H_2 - \frac{4}{3}\pi\lambda_1 d_{U_m}^3 \right) d(d_{U_m}). \end{aligned} \quad (15)$$

The influences of H_1 and H_2 on P_{suc} have different weights, which are proportional to λ_1 and λ_2 , respectively. According to (12), H_2 is mainly affected by the ratio of P_C to P_U . Hence, following (15), P_{suc} is mainly related with λ_1 , P_U , P_C , θ and α . Analyzing the performance on UAVs or CAs is a similar reasoning process, i.e., the SINR and successful received probability have similar analytic forms. The detailed difference between UAVs and CAs lies in the altitude, density and transmitting power of ADS-B. However, in this paper, our research mainly focuses on the analysis of the performance of GS on UAVs, and CAs are regarded as a interfering factor. Hence, we only provide the deductions and simulations concentrated on UAVs. Further, we analyze the detailed influences in Section IV.

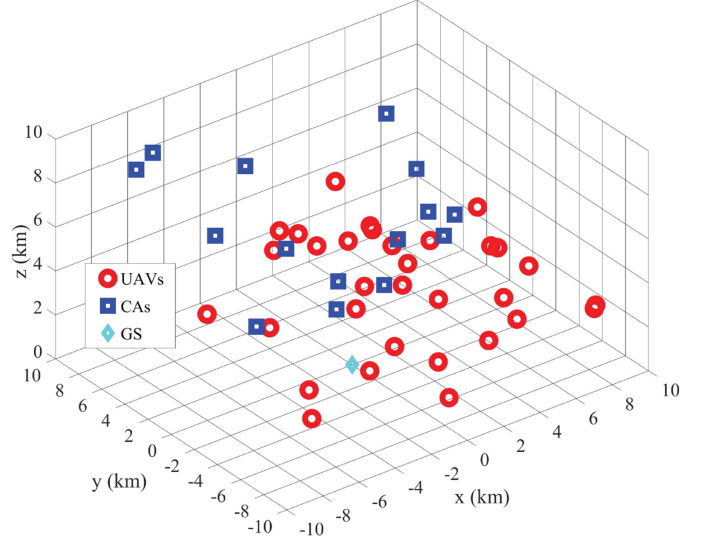


Fig. 2. Simulation scenario.

TABLE I
KEY PARAMETERS OF THE SIMULATION

Parameter	Value
B	1MHz
n_0	-174dBm/Hz
P_U	1W~70W
G_U	23dBi
P_C	15W~140W
G_C	20dBi
θ	7dB, 10dB, 11dB, 13dB, 14dB
α	2~5
λ_1	0~100
λ_2	15
H_U	1km~6km
H_C	6km~10km

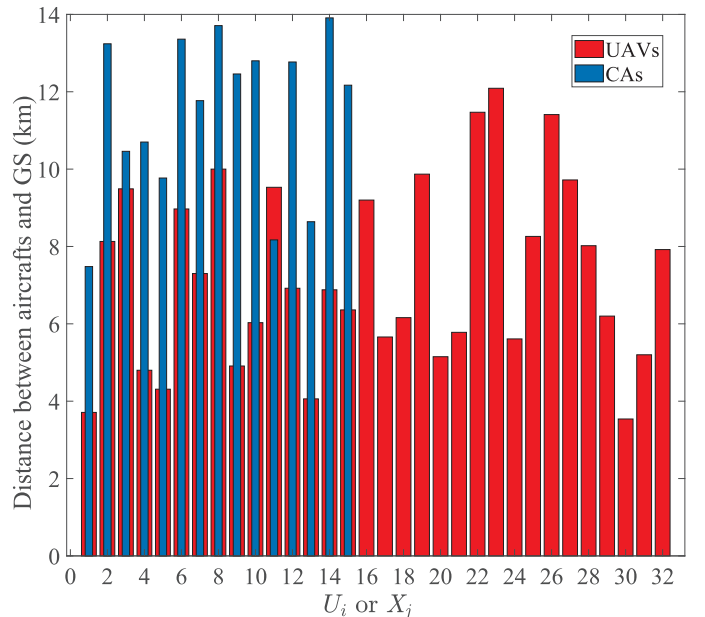


Fig. 3. Distance between GS and UAVs or CAs.

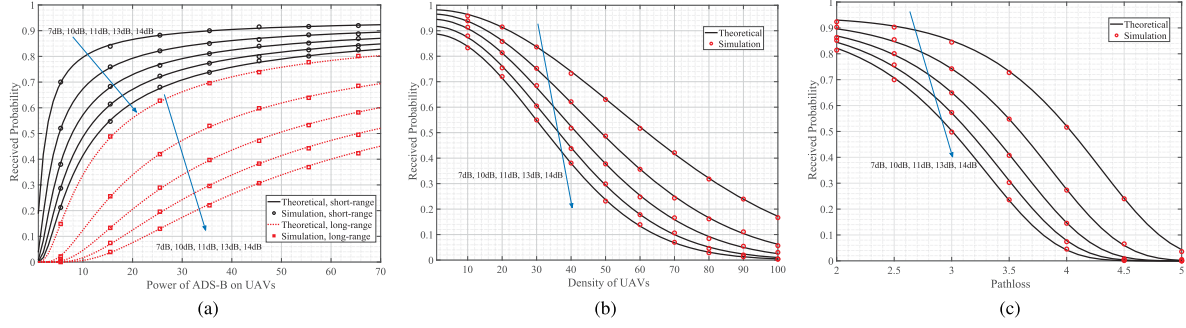


Fig. 4. Theoretical results and simulation results. (a) The impact of P_U on the received probability. (b) The impact of λ_1 on the received probability. (c) The impact of α on the received probability.

IV. SIMULATION RESULTS AND ANALYSES

To further evaluate the detailed performance, MATLAB is employed to simulate the 3D-PPP distribution scenario of the UAVs and CAs, as shown in Fig. 2. Space $\mathbf{V}$ is set as $20 \text{ km} \times 20 \text{ km} \times 10 \text{ km}$. The CAs are randomly distributed at altitudes H_C of $[6 \text{ km}, 10 \text{ km}]$, and the UAVs are randomly distributed at altitudes H_U of $[1 \text{ km}, 6 \text{ km}]$ due to the limited flight ability. λ_1 is set within $[1, 100]$, and λ_2 is fixed as 15. In the civil aviation system, the channel interval within $[962 \text{ MHz}, 1213 \text{ MHz}]$ is 1 MHz. Since we focus on ADS-B operating at 1090 MHz, the channel bandwidth B between the GS and all aircrafts are set as 1 MHz [16]. Besides, the pathloss index α is set within $[2, 5]$. Moreover, the total gain G_U on AG channel of UAVs is 23 dBi, and the total gain G_C on AG channel of CAs is 20 dBi. The detailed parameters in simulations are summarized in Table I.

Fig. 3 shows the distance between the GS and UAVs or CAs. In detail, X-axis represents the generated aircrafts, including UAVs and CAs, and Y-axis denotes the distance between the aircrafts and the GS, corresponding to the simulation scenario in Fig. 2. The distance of each point is randomly generated according to PPP. It is classified that if $d_{U_i} < 15 \text{ km}$, U_i is considered as a short-range UAV. Otherwise, if $d_{U_i} \geq 15 \text{ km}$, U_i is deemed as a long-range UAV.

Fig. 4(a) demonstrates the impact of P_U on the received probability under different threshold θ , considering UAVs as targets. P_C , α and λ_1 are respectively fixed at 30 W, 2 and 30. The solid lines represent the performance of short-range UAVs, while the dashed lines demonstrate the performance related to long-range UAVs. Besides, the simulation results coincide with the theoretical results, which validates the accuracy of theoretical analysis. As for the short-range UAVs, with the increment of P_U , the received probability from the desired UAVs at the GS increases. Moreover, as θ grows, the corresponding successful received probability decreases. When θ is 7 dB and P_U is 16 W, the received probability is 84.77%, where the received probability curve starts to smooth. If θ increases to 11 dB, the received probability descends to 68.63%. On the other hand, the dashed lines manifest the performance of the long-range flying UAVs. In particular, P_U has little influence on the received probability during the initial increment. Compared with short-range UAVs, with the increment of the distance, the received probability corresponding to the same θ and P_U declines. When θ is 7 dB and P_U is 16 W, the received probability is only 49.40%, which decreases by 41.72% on the basis of short-range UAVs. If the threshold escalates, more transmitting power is needed to compensate for the received probability. In conclusion, for short-range UAVs, setting P_U to be greater than 30 W is energy-consuming, since the received probability is already approaching flat. Furthermore, in terms of long-range UAVs, the received probability improves significantly when P_U falls between 30 W and 70 W [17]. Additionally, the curve of $\theta = 7 \text{ dB}$ and $\theta = 10 \text{ dB}$ show smooth trends when P_U exceeds 65 W.

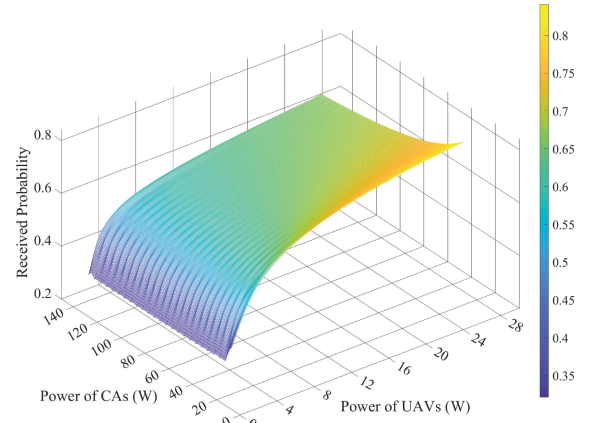


Fig. 5. The impact of P_U and P_C on the received probability.

To evaluate the density of UAVs, λ_1 in Fig. 4(b) is set within $[0, 100]$. We explore the impact of λ_1 on received probability under different threshold θ . P_U , P_C and α are respectively fixed at 11 dB, 30 W and 2. The simulation results coincide with the theoretical results, which validates the accuracy of theoretical analysis. As the density of UAVs ascends, the received probability descends. Supposing λ_1 is 30, as we set in Fig. 4(a), $\theta = 7 \text{ dB}$, $\theta = 10 \text{ dB}$, $\theta = 11 \text{ dB}$, $\theta = 13 \text{ dB}$, $\theta = 14 \text{ dB}$ correspond to the received probability of 83.68%, 75.05%, 67.69%, 60.61%, 54.86%, respectively. Besides, when θ is higher 11 dB and λ_1 exceeds 60, the GS can no longer support the monitoring of the airspace.

Fig. 4(c) discusses the impact of pathloss α on the received probability under different threshold θ . P_U , P_C and λ_1 are respectively fixed at 25 W, 30 W and 30. The simulation results coincide with the theoretical results, which validates the accuracy of theoretical analysis. Primarily, α has little influence on the received probability during the initial increment. When α exceeds 3, the received probability drops sharply. Considering θ is 7 dB and α is 3, the corresponding received probability is 85.1%. When α rises to 4.5, the corresponding received probability declines to 24.1%. If θ increases at this point, the GS can no longer support the monitoring of the airspace. In short, when α is greater 3, the increment cause the received probability to plummet, making the channel performance deteriorated. By increasing P_U or decreasing θ , we can compensate for the effect of the increment of α on the received probability.

Fig. 5 illustrates the impact of P_U and P_C on received probability. We set P_C and P_U as variables, aiming to simultaneously examines the influence of P_U and P_C on the received probability at short-range UAVs. θ , α and λ_1 are respectively fixed at 7 dB, 2 and 30. With the increment of P_U , the received probability increases. On the contrary, as

P_C enlarges, the received probability diminishes. Assuming P_U is 24 W and P_C is 40 W, the corresponding received probability is 75.26%. If P_C stays constant, the corresponding received probability is 70.98% when P_U decreases to 15 W. If P_U stays constant, the corresponding received probability is 70.49% when P_C increases to 73 W. Minishing P_C improves the received probability of the GS towards UAVs. However, the monitoring performance of the GS towards CAs is undermined. In addition, P_U is unadvisable to be magnified indefinitely, which intensifies the signal interference, impairs the ability of the GS to monitor CAs and wastes energy. Hence, the transmitting power of both types of aircrafts should be balanced according to actual demands.

V. CONCLUSION

This work analyzes the interference for the coexistence of UAVs and CAs based on ADS-B. We build a 3D-PPP model and deduces the explicit analytic form of the received probability targeting UAVs by SG theory. When analyzing the signal of a UAV, the interference signals are distinguished between UAVs and CAs. Moreover, based on the AG channel, we reveal the effects of transmitting power, density, threshold and pathloss on the performance of the ADS-B monitoring system via simulations. Additionally, the transmitting power of UAVs and CAs are both taken as variables to analyze the received probability of UAVs. In terms of raising the received probability, the transmitting power of ADS-B and the density of UAVs are contrarious, which can be set appropriately according to the requirements of the surveillance performance. In short, this work contributes to the appropriate deployment of ADS-B equipment on the UAVs, which helps improve the airspace safety and enhances the air traffic flow management.

REFERENCES

- [1] H. Xu et al., "A survey on UAV applications in smart city management: Challenges, advances, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 8982–9010, 2023.
- [2] L. Gupta, R. Jain, and G. Vaszkun, "Survey of important issues in UAV communication networks," *IEEE Commun. Surveys Tut.*, vol. 18, no. 2, pp. 1123–1152, Secondquarter 2016.
- [3] Y. Zhu et al., "UAV trajectory tracking via RNN-enhanced IMM-KF with ADS-B data," in *Proc. IEEE Wireless Commun. Netw. Conf.*, Apr. 2024, pp. 1–6.
- [4] M. Strohmeier, V. Lenders, and I. Martinovic, "On the security of the automatic dependent surveillance-broadcast protocol," *IEEE Commun. Surveys Tut.*, vol. 17, no. 2, pp. 1066–1087, Secondquarter 2015.
- [5] M. Strohmeier, M. Schafer, V. Lenders, and I. Martinovic, "Realities and challenges of nextgen air traffic management: The case of ADS-B," *IEEE Commun. Mag.*, vol. 52, no. 5, pp. 111–118, May 2014.
- [6] X. Deng, F. Wang, and G. Yang, "A survey on airborne ADS-B technology and its development trend," *Adv. Aeronautical Sci. Eng.*, vol. 12, no. 1, pp. 121–128, Feb. 2021.
- [7] P. Jonas, M. Jancik, S. Holoda, and J. Bodart, "Impact of SUAS equipped with ADS-B on 1090 MHz environment," in *Proc. New Trends Civil Aviation*, 2020, pp. 63–67.
- [8] H. Liu, S. Wu, D. Qin, and D. Li, "Performance analysis of surveillance capacity of satellite-based ADS-B receiver," *Acta Aeronautica Et Astronautica Sinica*, vol. 39, no. 5, pp. 8–15, May 2018.
- [9] J. G. Andrews, F. Baccelli, and R. K. Ganti, "A tractable approach to coverage and rate in cellular networks," *IEEE Trans. Commun.*, vol. 59, no. 11, pp. 3122–3134, Nov. 2011.
- [10] H.-S. Jo, Y. J. Sang, P. Xia, and J. G. Andrews, "Heterogeneous cellular networks with flexible cell association: A comprehensive downlink SINR analysis," *IEEE Trans. Wireless Commun.*, vol. 11, no. 10, pp. 3484–3495, Oct. 2012.
- [11] L. Zhou, Z. Yang, S. Zhou, and W. Zhang, "Coverage probability analysis of UAV cellular networks in urban environments," in *Proc. IEEE Int. Conf. Commun. Workshops*, 2018, pp. 1–6.
- [12] Q. Zhang, J. Chen, L. Ji, Z. Feng, and Z. Chen, "Response delay optimization in mobile edge computing enabled UAV swarm," *IEEE Trans. Veh. Technol.*, vol. 69, no. 99, pp. 3280–3295, Mar. 2020.
- [13] X. Liu, "Closed-form coverage probability in cellular networks with poisson point process," *IEEE Trans. Veh. Technol.*, vol. 68, no. 8, pp. 8206–8209, Aug. 2019.
- [14] Y. Guo, X. Jia, S. Cao, and Z. Hao, "Analysis of downlink coverage and capacity for 3D mobile UAV networks," in *Proc. IEEE 7th Int. Symp. Mechatron. Ind. Inform.*, 2021, pp. 236–239.
- [15] H. E. Sawy, A. Sultan-Salem, M. -S. Alouini, and M. Z. Win, "Modeling and analysis of cellular networks using stochastic geometry: A tutorial," *IEEE Commun. Surveys Tut.*, vol. 19, no. 1, pp. 167–203, Firstquarter 2017.
- [16] Y. Bai, X. Zhang, and L. Zhao, "Co-channel CCK transmission overlapped with DME in aeronautical communication," in *Proc. IEEE China Summit Int. Conf. Signal Inf. Process.*, Xi'an, China, 2014, pp. 403–406.
- [17] RTCA, "Minimum operational performance standards (MOPS) for 1090MHz extended squitter automatic dependent surveillance-broadcast (ADS-B) and traffic information services-broadcast (TIS-B)," 2010. [Online]. Available: <https://www.rtca.org/standards/>